FORMULA SAE®

UNICAMP RACING

Design Briefing

Universidade Estadual de Campinas

Unicamp E-Racing #E08

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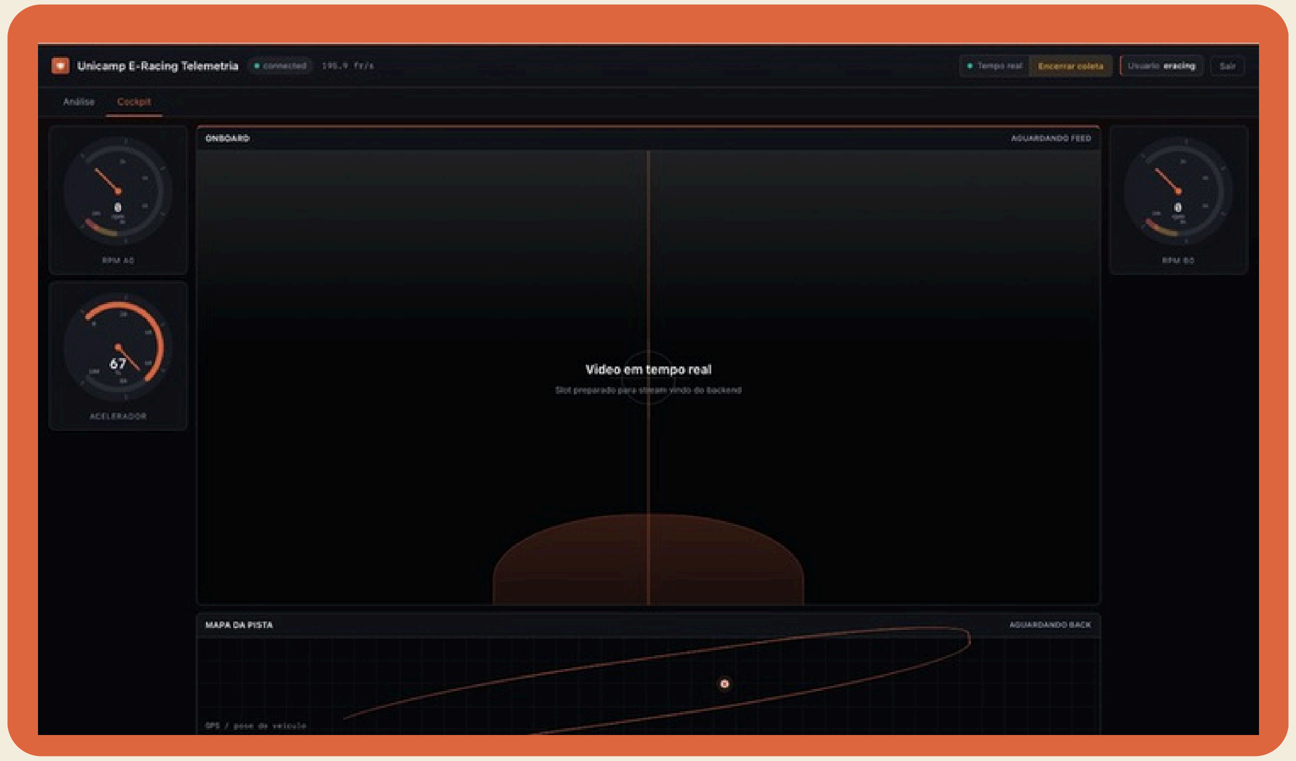
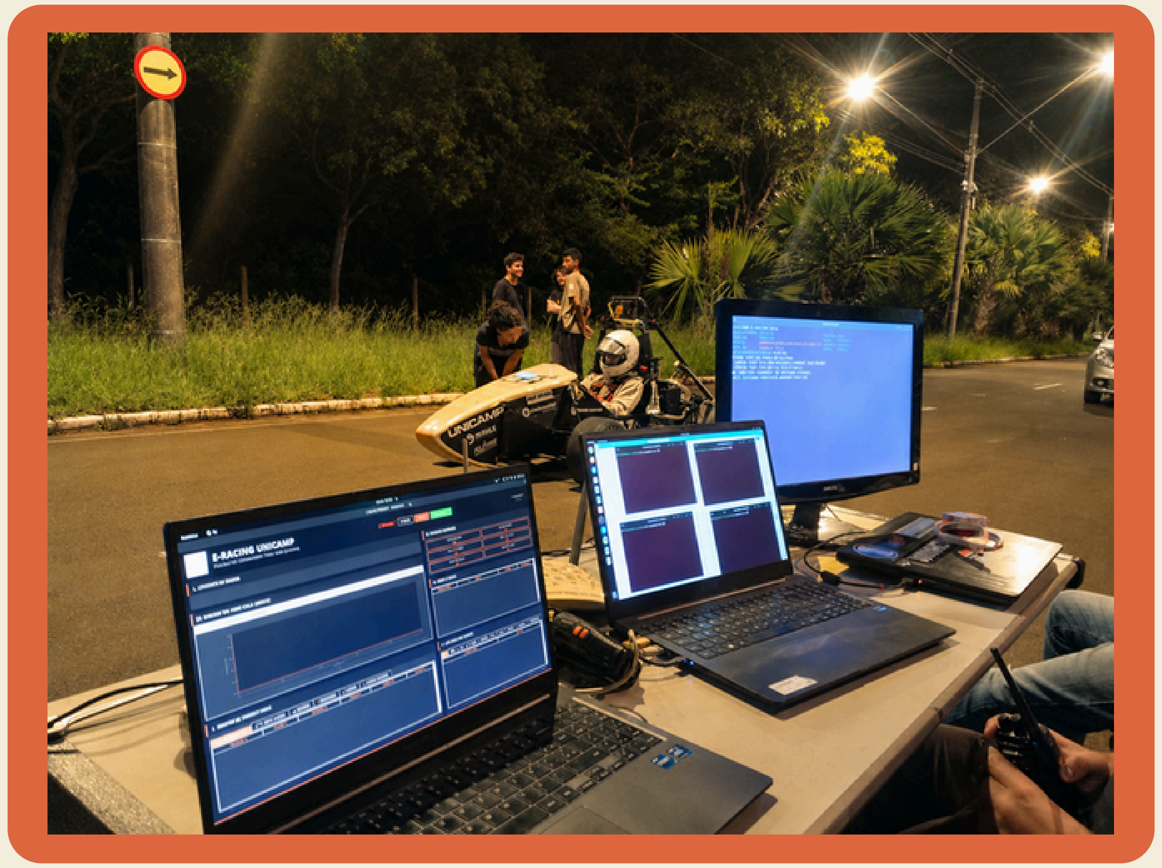


Cross-Session Performance Comparison

- Consistent log structure allows comparison across different test days and track conditions
- Signal behavior trends (thermal rise, voltage sag, power asymmetry) trackable over time
- Data supports setup decisions between sessions



Post-Session Data Analysis

- Full session logs stored and accessible after each run
- RPM-relative historical view enables lap-by-lap performance breakdown
- Engineers can isolate specific time windows of the track for granular signal inspection
- Selected window plots min / avg / max per signal for direct comparison between sessions or laps

Real-Time Safety Monitoring

- Dashboard displays critical vehicle safety parameters live during and after sessions
- Engineers monitor min / avg / max values per signal directly from the stats bar.

Multi-Signal Plotting

- Multiple signals overlaid simultaneously
- Visual correlation between signals allows identification of thermal, electrical and mechanical anomalies
- Window selection tool enables focus on critical moments braking zones, acceleration peaks, corner exits

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Telemetry

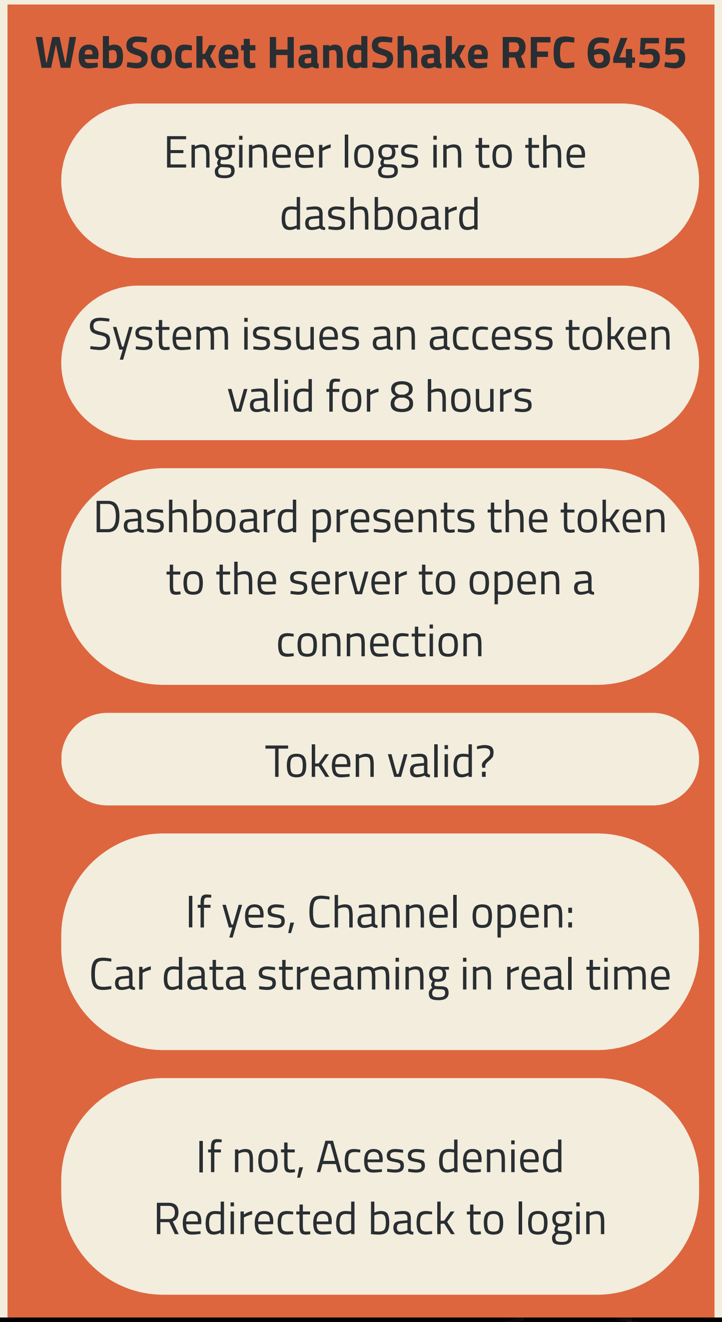
Integrated Infrastructure

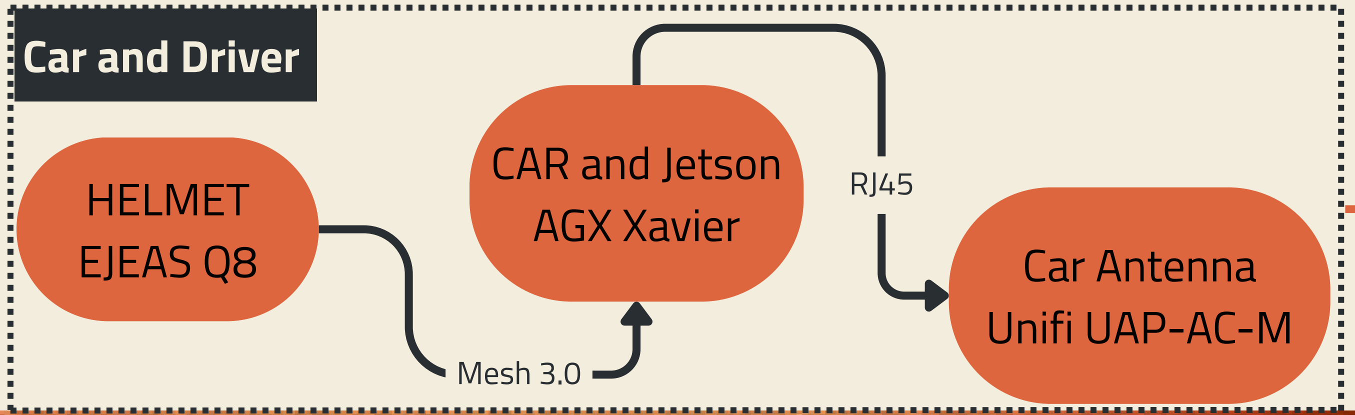
- Data Acquisition
  - Real-time CAN bus monitoring with 38 signals at 500 kbps
  - Rust pipeline: edge processor → server → live dashboard
  - Dual persistence: time-series database + permanent storage
  - Autonomous boot and automatic reconnection
- Video Streaming
  - Stereo camera stream via low-latency UDP H.264
  - Centralized relay accessible from dashboard and mobile app
  - Automated session backup: 5 minute MKV segments
- Pilot Communication
  - Full-duplex wireless system (dedicated 2.4 GHz channel)
  - Completely isolated from telemetry Wi-Fi band (5 GHz)
  - Full session logging and audio backup
- Security & QoS
  - JWT authentication on all endpoints (8-hour tokens)
  - Zero Trust architecture for remote access
  - Structured QoS: telemetry and audio always prioritized over video
  - bcrypt password hashing: brute-force resistant by design

Metric	Measured (Bench)	Target
CAN→Screen Latency	3.77ms	<50ms
Throughput	1000+ msg/s	> 500msg/s
Unifi UAP-AC-M Range	~50 m (Indor)	> 200 m
Pilot→Pit Audio Latency	70 ms	< 150 ms
UDP/RTSP Video Latency	~80 ms	< 500 ms
Total Bandwidth Usage	~6.5 Mbit/s	< 50 Mbit/s
Jetson↔Server Clock Offset	±0.1 ms post-NTP	±1 ms
Boot-to-Data Time	51 s (Jetson)	< 120 s

Blue Team: Monitoring and Defense

- Active Monitoring
  - Real-time process-level bandwidth tracking and centralized log analysis.
  - Continuous JWT auditing and unauthorized device detection via RSSI.
- Network Defense
  - Strict firewall policies, anti-DDoS measures, and automated brute-force protection.
  - Global access secured via Cloudflare Zero Trust (Identity-aware SSH).
- Application Security
  - Encrypted traffic (WSS/HTTPS) and structural immunity against SQL injection.
- Incident Response
  - Tools for immediate attacker isolation and global session revocation.





Telemetry-Project Objectives

Engineering Mission:

- Data Acquisition:
  - Real-time CAN bus monitoring with 38 signals at 500 kbps
  - Rust pipeline: edge processor → server → live dashboard
  - Dual persistence: time-series database + permanent storage
  - Autonomous boot and automatic reconnection
- Real Time Security:
  - Continuous monitoring of critical parameters, such as maximum cell temperature, VCU status, and minimum cell voltage.
  - System designed for low latency, ensuring that readings are maintained virtually in real time.
- Historical foundation for continuous development:
  - Accumulate structured logs from each testing session in a persistent database, building a technical memory of the vehicle's behavior that informs the development of the next prototype

Contribution to the Team's Overall Objectives:

Team Guiding Principle	Telemetry Feedback and Response
Conservative Approach	Prioritized reliability and operational simplicity. Fully open-source stack with no external vendor dependency.
Cost Reduction	Open-source stack (Rust, TimescaleDB, SQLite, SolidJS, MediaMTX) with no licensing costs. Eliminates the need for a commercial DAQ.
Safety Focus	Real-time monitoring of the BMS, inverters, and VCU with 1–2 ms latency. Remote emergency stop through the dashboard.

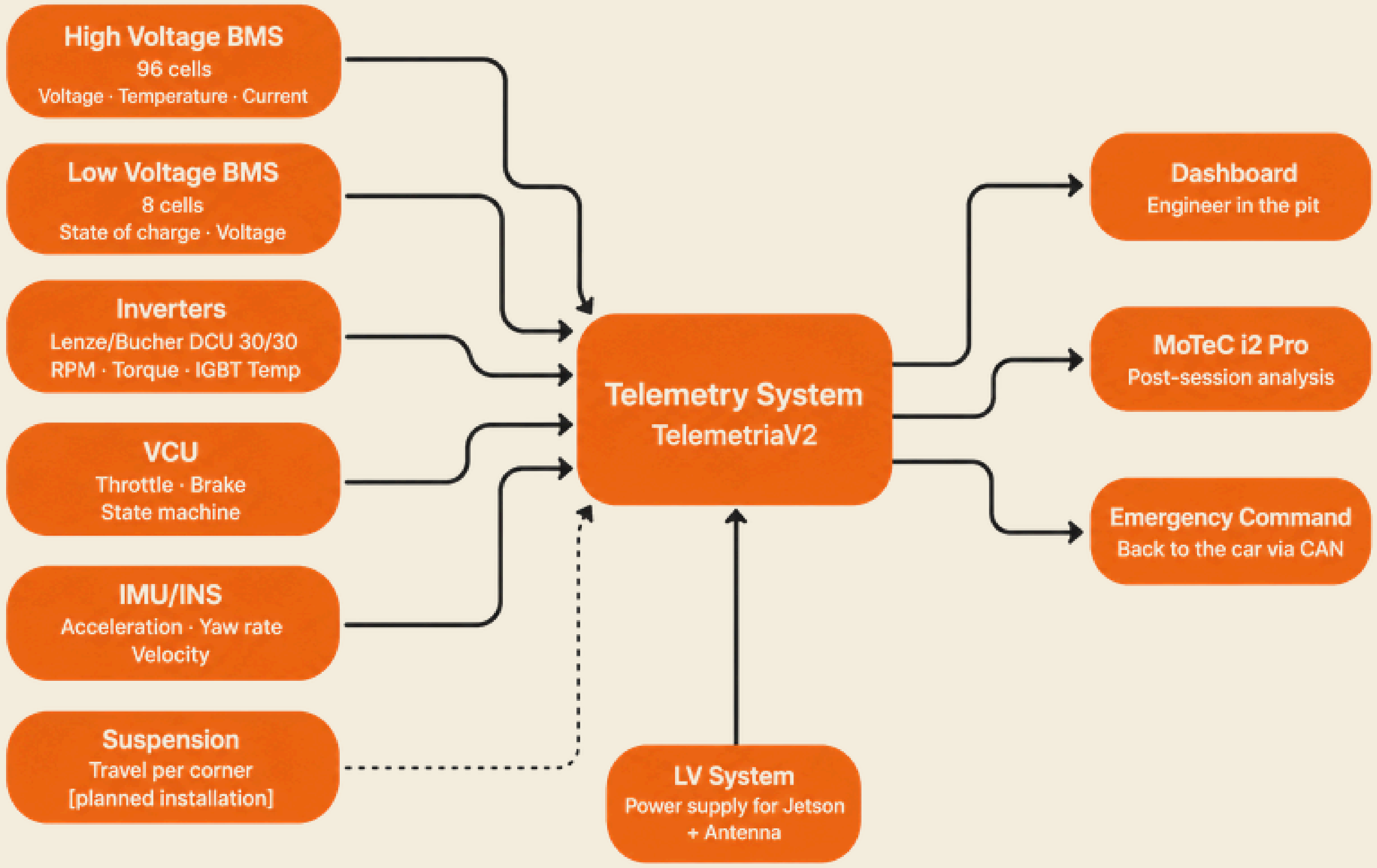
Telemetry-Project Objectives

Key Contributions to Vehicle Performance

Telemetry does not directly modify the vehicle during a race, but it determines how quickly the team can understand what happened and make the next decision. This influence manifests itself at three levels:

- Safety Level:
  - Real-time visibility of critical parameters enables intervention before failures occur. Inverter temperature spikes or sudden cell voltage drops appear on the dashboard in under 2 ms, allowing rapid response.
- Setup Level:
  - Post-session analysis in MoTeC enables comparison of suspension, torque distribution, and braking configurations across laps, supporting faster setup optimization.
- Development Level:
  - Accumulated logs create a historical database of vehicle behavior, supporting future prototype development, thermal model validation, and regenerative system studies.

Integration with Other Subsystems



LV Dependency:

Telemetry relies on the LV system for power. Any LV failure results in a complete telemetry outage, making LV stability essential for data system reliability.

Trade-offs and Design Decisions

- Range vs. Latency:
  - Lower latency requires positioning the base station close to the track. Direct line-of-sight minimizes retransmissions and latency spikes.
- Complexity vs. Robustness:
  - V2 adds complexity (async Rust, dual databases, JWT, QoS) to solve V1 limitations. Reliability is maintained through local buffering and automatic reconnection
- Dual CAN Buses vs. Simpler Harness:
  - Using can0 and can1 increases harness complexity but prevents inverter traffic from overloading the shared bus and affecting other subsystems.

Telemetry-Project Design

System Requirements and Their Definition:

- FSAE Rules:
  - Electrical isolation, fuse protection, and secure Jetson mounting within the vehicle.

- Limitations identified in V1:
  - Growing latency and the absence of persistent data storage—defined the two most important technical requirements for V2: stable latency regardless of session duration and dual storage (TimescaleDB for real-time data, SQLite for permanent history).

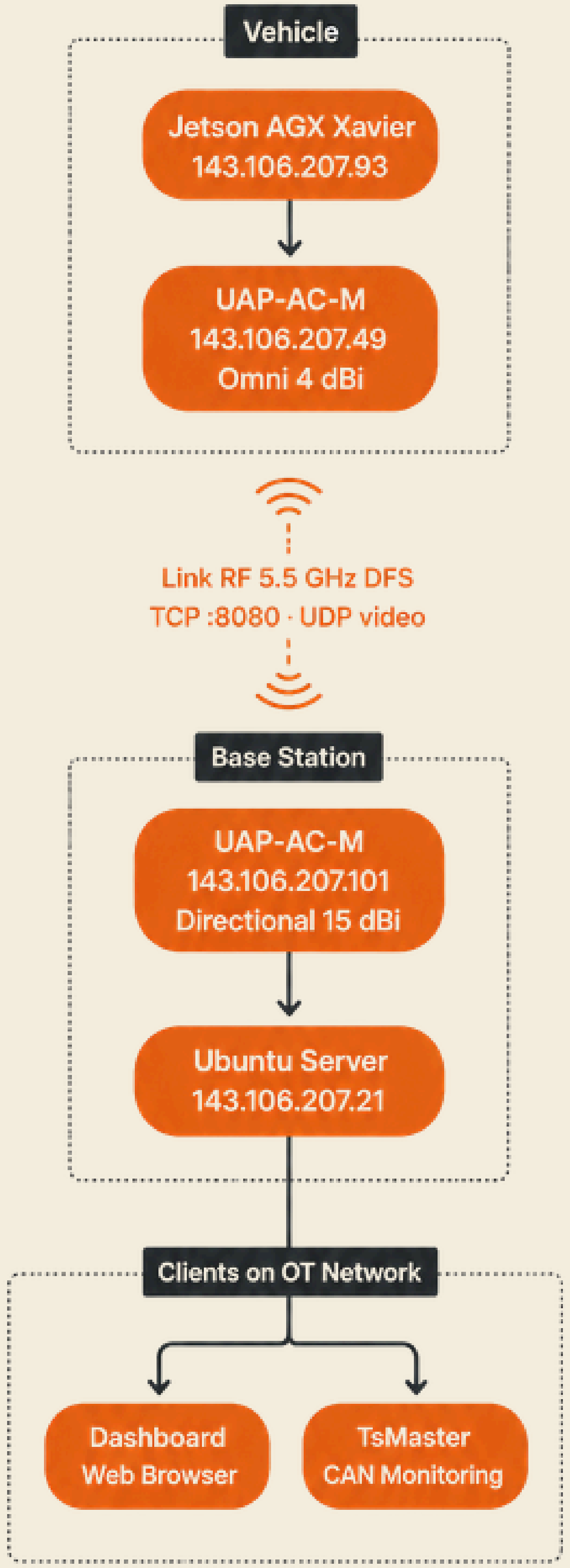
- From the subareas:

: Subarea	Requested Requirement
Dynamics / Suspension	Suspension travel at 100 Hz, correlated with IMU data
Powertrain	Motor and inverter temperatures with sufficient resolution for thermal model validation
Electrical	Real-time monitoring of all 96 HV BMS voltage and temperature channels
Safety	Maximum acceptable monitoring latency: < 100 ms

Network Architecture and Topology:

Device	Static IP	Function
Ubuntu Server	143.106.207.21	Data ingestion, database, and dashboard hosting
Jetson AGX Xavier	143.106.207.93	Edge node and CAN data acquisition
Pit AP (Directional)	143.106.207.101	Pit lane access point
Vehicle AP (Omnil)	143.106.207.49	Vehicle access point

The network follows an isolated Operational Technology (OT) architecture, with no internet connection during operation. The Ubuntu server handles all network services, including DHCP, internal DNS, and request routing, while the physical router operates in bridge mode, providing only the communication medium.



Telemetry-Project Design

Physical RF Link Parameters

The RF link operates at 5.5 GHz using DFS channels, avoiding the severe 2.4 GHz congestion typically found at FSAE events with thousands of spectators and dozens of teams.

- Equipment:
  - Vehicle (TX):
    - Ubiquiti UAP-AC-M with omnidirectional toroidal antenna
  - Pit Station (RX):
    - Ubiquiti UAP-AC-M with directional antenna

Parameter	Value
TX Power	** a ser confirmado no datasheet
Omnidirectional Antenna Gain	
Directional Antenna Gain	
RX Sensitivity (MCS7)	
Estimated Cable Loss	1.0 dB

Link Budget Modeling (Friis Equation)

Free-space path loss (FSPL) follows the Friis equation:

FSPL (dB)=32.45+20log10(Dkm)+20log10(fMHz)FSPL\,(dB) = 32.45 + 20\log_{10}(D_{km}) + 20\log_{10}(f_{MHz})FSPL(dB)=32.45+20log10(Dkm)+20log10(fMHz)

24.5 dB fade margin at 900 m — well above the 15–20 dB design target.

Distance	FSPL	RX Power	Fade Margin
50m	83.5 dB	-45.5 dBm	44.5 dB
150m	93.0 dB	-55.0 dBm	35.0 dB
300m	96.8 dB	-58.8 dBm	31.2 dB
600m	102.8 dB	-64.8 dBm	25.2 dB
900m	103.5 dB	-65.5 dBm	24.5 dB

All tested distances maintain a fade margin above 20 dB, ensuring reliable communication throughout the competition track

Telemetry-Project Design

Channel Capacity (Shannon–Hartley Theorem):

C = BW * log2(1 + SNR)

C = 20 * 10^6 * log2(1 + 4168)

C ≈ 240.4 Mbps

Actual System Load:

- CAN Telemetry (TCP): ~240 kbps
- H.264 Video Stream (UDP): ~4 Mbps
- Total Traffic: ~4.24 Mbps

Custom Binary Protocol — Frame Format:

Each transmitted frame by the network has 20 btyes

Transport Protocol Comparison:

Criteria	TCP	UDP
Data Integrity	Guaranteed	Not guaranteed
Typical Latence	1-2 ms	< 1 ms
Vehicle Telemetry	Recommended	Not suitable
Video Streaming	Unnecessary	Recommende d

CAN → Server (TCP vs. UDP)

TCP Selection: Frame loss compromises vehicle dynamics analysis and MoTeC post-processing, making TCP the preferred choice despite the slightly higher latency.

Criteria	Binary WebSocket	HTTP Polling
Latency	< 2 ms	> 50 ms
Overhead	Low-Single Handshake	High-Repetead Handshake
Real Time Telemetry	Ideal	Not viable

Dashboard Communication (WebSocket vs. HTTP Polling)

WebSocket Selection: Instant push updates and low communication overhead enable reliable real-time telemetry monitoring.

Telemetry-Project Design

Clock Synchronization — Chrony as the Final Solution:

Accurate clock synchronization between the Jetson and the server is essential for reliable telemetry analysis. Even small clock offsets can misalign CAN events and vehicle responses, compromising data correlation in MoTeC.

Issue Identified During Testing:

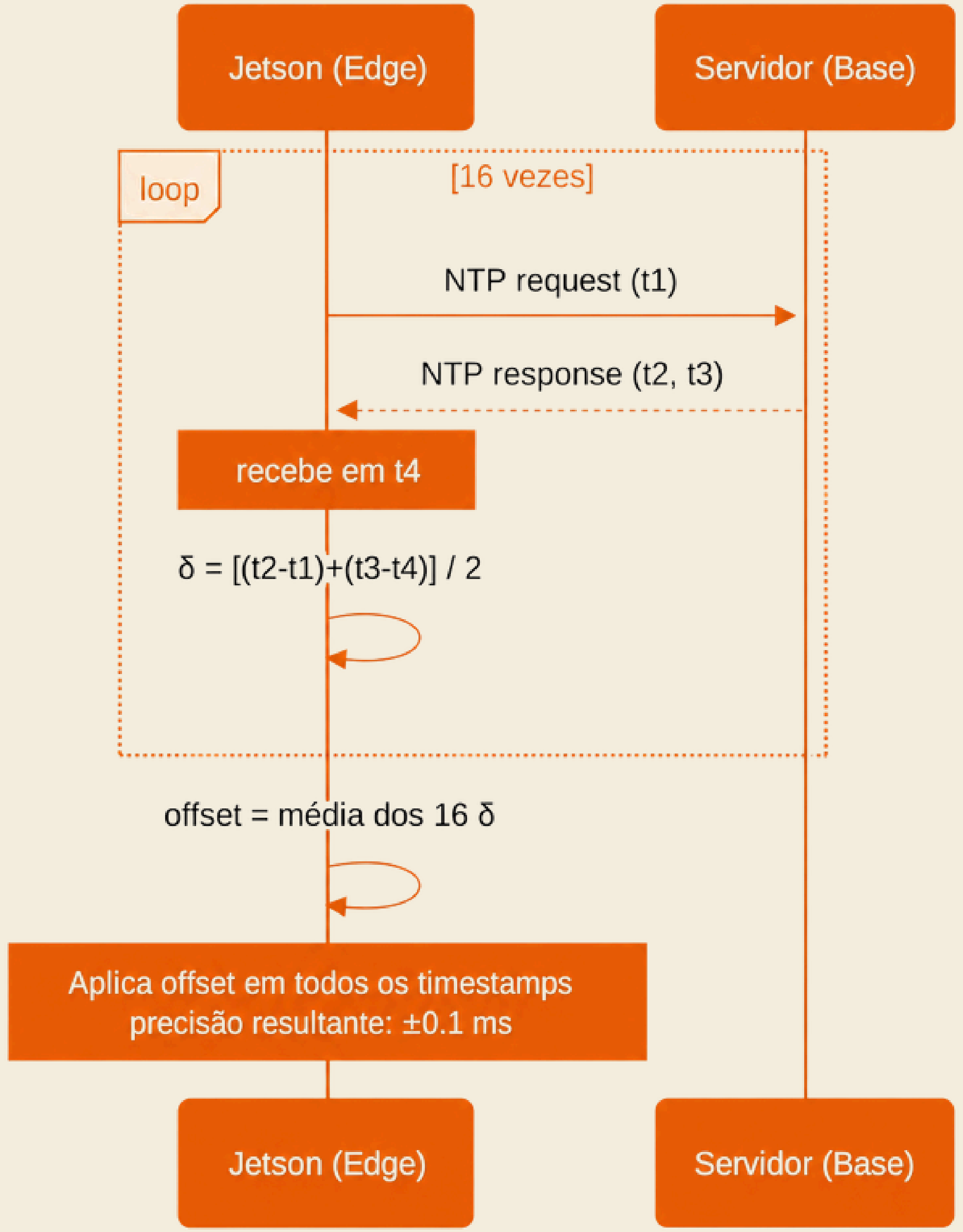
Measured latency appeared to increase from ~14.5 ms to over 60 ms during long sessions. Investigation showed that the cause was not network degradation, but clock drift between the Jetson and the server.

Limitation of the Previous Application:

The original system used a custom NTP synchronization performed only once at startup. Because quartz oscillators continuously drift over time, synchronization error accumulated throughout the session, creating the illusion of increasing network latency

Chrony as solution:

Chrony was adopted as the operating system time manager, continuously correcting clock drift in real time.



The application's internal NTP was disabled (--ntp-port 0 in the Jetson service), delegating all synchronization responsibilities to Chrony. As a result:

- Accumulated drift is eliminated — Chrony continuously corrects the clock.
- Frame timestamps reflect the server's actual clock within < 1 ms.
- Dashboard latency remains stable throughout the entire session.

Measured Result:

Stable latency between 1–49 ms across complete sessions, with no temporal growth.

Configuration Persistence on the Jetson:

A legacy NetworkManager script (99-eracing-route.sh) overwrote network settings at every boot. /etc/resolv.conf was also overwritten by systemd-resolved. The solution consisted of a three-layer approach

Telemetry-Project Design

Antenna Configuration and Justification: Directional vs. Omnidirectional

The choice of the physical arrangement and antenna types on the vehicle and at the pit station directly reflects the vehicle's motion dynamics and the track geometry:

- On the Vehicle (UAP-AC-M + Standard Omni Antennas):
  - Gain: 4 dBi at 5 GHz.
  - Radiation Pattern: Toroidal (uniform 360° coverage in the horizontal plane).
  - Justification: The vehicle constantly changes orientation, making omnidirectional coverage essential for maintaining a reliable link.
- At the Pit Station (UAP-AC-M + UMA-D Directional Panel Antenna):
  - Gain: 15 dBi at 5 GHz (10 dBi at 2.4 GHz).
  - Beamwidth: 45° horizontal and vertical.
  - Justification: As a fixed monitoring point, the pit station concentrates RF energy toward the track, increasing gain by approximately 11 dB and enabling reliable communication over long distances while reducing free-space path loss.
  - Isolation and Polarization: Dual-linear polarization with 30 dB cross-polarization isolation. This reduces polarization mismatch caused by vehicle movement and improves 2×2 MIMO performance.

Synchronous Processing and Resampling (ZOH) at 100 Hz:

The CAN network operates inherently asynchronously, with messages transmitted either by events or at different frequencies (inverters at 100 Hz, VCU at 50 Hz, BMS at 10 Hz). However, the professional MoTeC .ld telemetry format requires every channel to have a fixed synchronous sampling rate over time (an implicit time axis derived from a single logging frequency).

TelemetryV2 Approach (implemented in Rust within logs.rs):

- The backend intercepts CAN frames and performs synchronous resampling onto a fixed 100 Hz grid (LD_SAMPLE_RATE_HZ).
- The interpolation method used is Zero-Order Hold (ZOH). This method assumes that a physical signal remains constant at its last measured value until a new CAN message updates its state.
- ZOH accurately models the behavior of discrete CAN buses without introducing artificial noise. It ensures that asynchronous signals from different subsystems remain aligned millisecond by millisecond in the output file, enabling high-integrity overlays and dynamic calculations in MoTeC i2 Pro.

Telemetry-Project Validation

Experimental Setup:

- **Track 1 (Autocross/Endurance/Skid Pad):**
  - Maximum distance from the base station: ~150 m
  - Base station positioned on a fixed platform with direct line-of-sight to the track
  - Condition: vehicle in motion during normal testing sessions
  - Active sensors: HV BMS, LV BMS, inverters, VCU, and IMU
- **Track 2 (Acceleration Straigh):**
  - Maximum distance from the base station: ~300 m
  - Condition: vehicle in motion
  - Active sensors: same as Track 1
  - Note: Suspension travel sensors were not installed during these sessions.

RF Propagation and L2 Efficiency Correlation:

Distance	Theoretical RSSI (Friis)	Measured RSSI	Estimated SNR	PDR (%)	PER (%)
50m	−43.2 dBm	−28.7 dBm	-	-	-
150m	−52.7 dBm	−28.7 dBm	-	-	-
300m	−58.8 dBm	−65.3 dBm	-	-	-

Measured RSSI obtained through the Ubiquiti UniFi Controller, with the vehicle in motion

Latency and Throughput

Condition	RSSI	Min Latency	Max Latency	Latency (Avg.)	Throughput
Track 1 (≤150 m)	−28.7 dBm	1 ms	39 ms	20 ms	1,112 fps
Track 2 (≤300 m)	−65.3 dBm	1 ms	49 ms	20 ms	1,112 fps
~200 m (estimated)	(~−48 dBm)	-	-	20 ms	-

Latency measured as the difference between the frame timestamp (OS clock synchronized by Chrony) and the reception time at the server. Stable throughout the entire session after Chrony implementation.

Telemetry-Project Validation

Comparation V1 vs V2

Metric	V1	V2
Initial Latency	~2 ms	1-2 ms
Latency After 3 min	~1.5 min	1-49 ms (stable)
Sustained Throughput	Not measurable	1.112 fps
Frame Loss Due to Degradation	Yes	Not observed
Accumulated Clock Drift	Present (fixed offset)	Eliminated (chrony sync)
Video Support	No	Yes
Emergency Command	No	Kill Resume validated on track
Historical Data Storage	Session CSVs	TimescaleDB + SQLite

Track Validation

First test with the real vehicle. Results:

Validated Item	Result
VCU signals	Displayed on the dashboard in real time
Inverter signals	Displayed on the dashboard in real time
Emergency Stop via dashboard	Motor shut down when KILL was pressed
Emergency Resume via dashboard	Vehicle restarted when RESUME was pressed
Correct CAN frame (ExtendedId, zero payload)	Confirmed via candump
Kill command on can0 and can1	VCU responded correctly
Stable latency during the session	No growth after Chrony implementation

Post-Session Analysis in MoTeC i2 Pro

The .ld and .ldx files generated by the server were validated in MoTeC i2 Pro, with all active channels displaying correct physical values and properly aligned timestamps.

Setup Analysis Example:

The comparison of front suspension travel against lateral acceleration within a specific speed range allows identification of the vehicle's roll behavior in corners. If the suspension trace shows excessive compression on the outside front corner without a proportional response at the rear, this indicates structural understeer. A front spring adjustment (from setup A to setup B) would be decided based on this comparison and validated in the following session.

Security and Network Monitoring:

The dashboard admin panel provides real-time visibility into the following metrics:

Metric	Source	Frequency
CAN → Server Latency	Server atomic counter	Per frame
Frame Rate (fps)	Atomic counter, computed every 10 s	Every 10 s
HTB Class Bytes/ Packets	tc -s class show dev <iface>	3 s polling
Antenna RSSI	Ubiquiti UniFi Controller API	
Active WebSocket Connections	Server broadcast channel	Real time

Telemetry-Future Projects

Competition Validation at 900 m — FSAE Brazil 2026

The July 2026 competition will be the system's first full-scale deployment, with distances of up to 900 m between the vehicle and the base station. The Friis model predicts PRX = –65.5 dBm at 900 m, with a 24.5 dB fade margin, exceeding the conservative 15–20 dB threshold.

RSSI and latency will be continuously logged through the UniFi Controller API and correlated with vehicle position to validate the theoretical model under real competition conditions.

RSSI Integration as a Synthetic CAN Signal

A planned extension of the edge agent will read RSSI values using iw dev wlan0 link at 1 Hz and inject them as a synthetic signal into the CAN pipeline. This will enable RF link quality overlays in MoTeC i2 Pro alongside position and lap-time data.

QoS and Simultaneous Video Streaming

H.264 video streaming through GStreamer and MediaMTX has been implemented but has not yet been tested simultaneously with CAN telemetry. The next session will evaluate HTB scheduler performance under combined load (~4.24 Mbps), measuring CAN frame latency impact and video jitter.

Multi-Antenna Network with Graph-Based Selection Algorithm

The current system uses a single antenna pair (vehicle ↔ pit station). On tracks with complex geometry or distances exceeding 600 m, a single directional antenna may struggle to maintain line-of-sight throughout the entire course.

The proposed solution distributes multiple fixed antennas around the track, connected to the server via wired infrastructure. A graph- and RSSI-based selection algorithm dynamically chooses the best antenna for the vehicle at each position.

1.8. **Low Voltage Battery**

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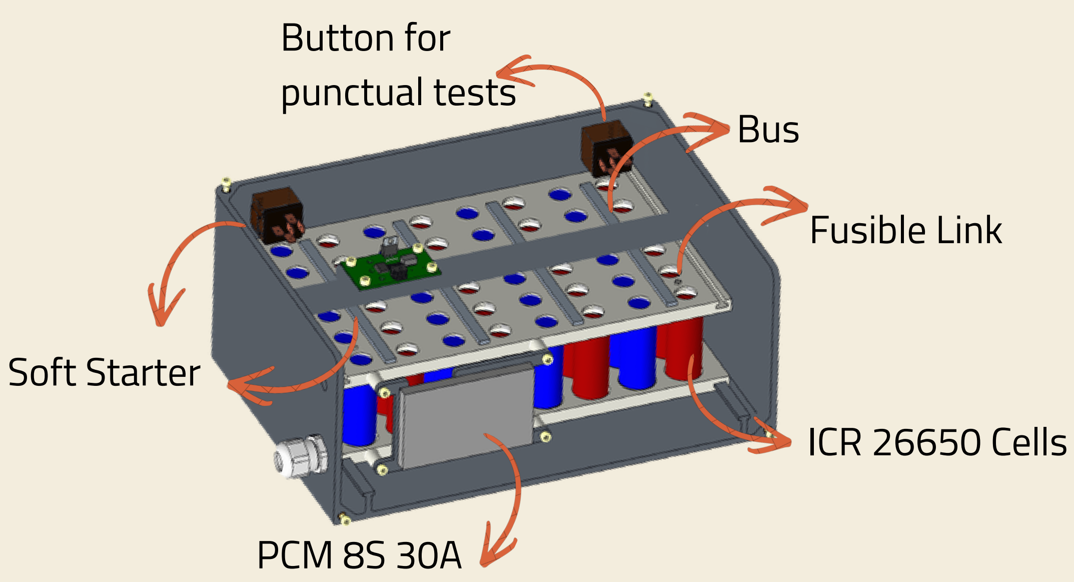
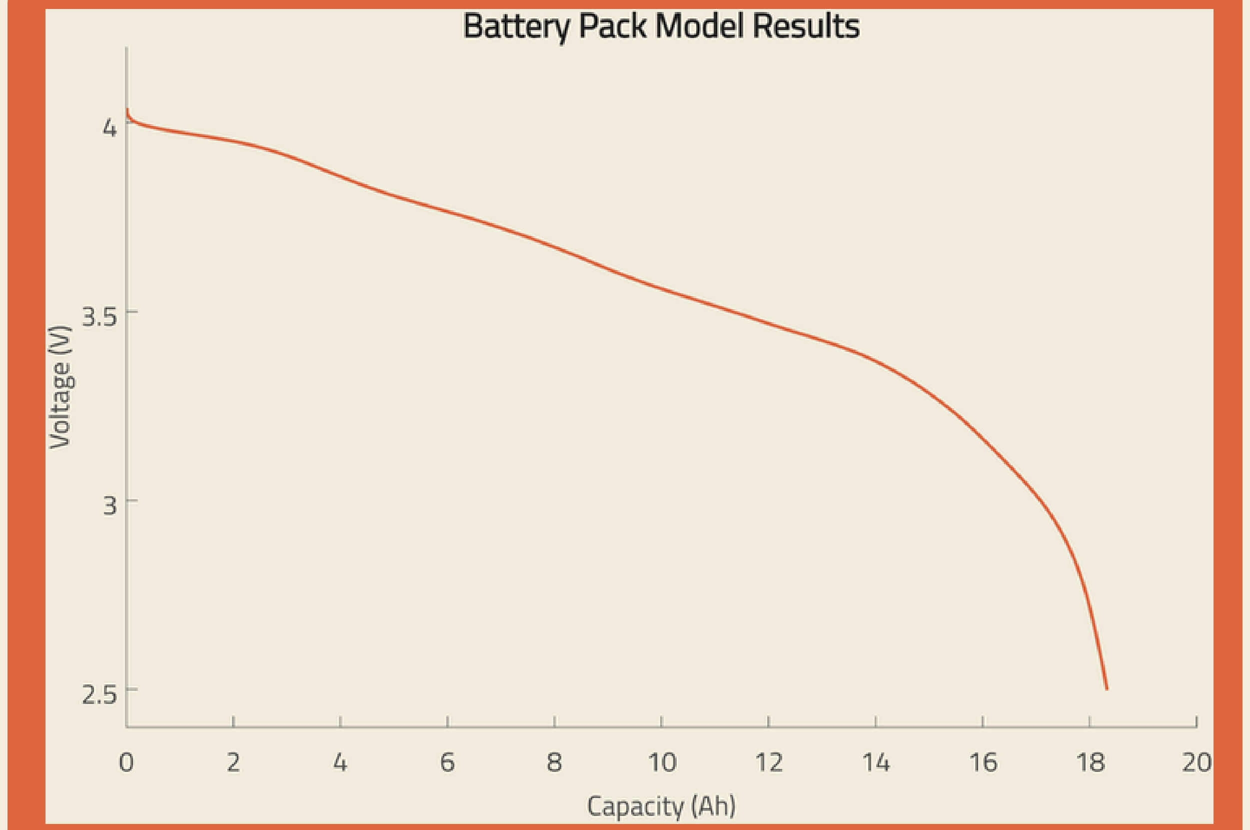
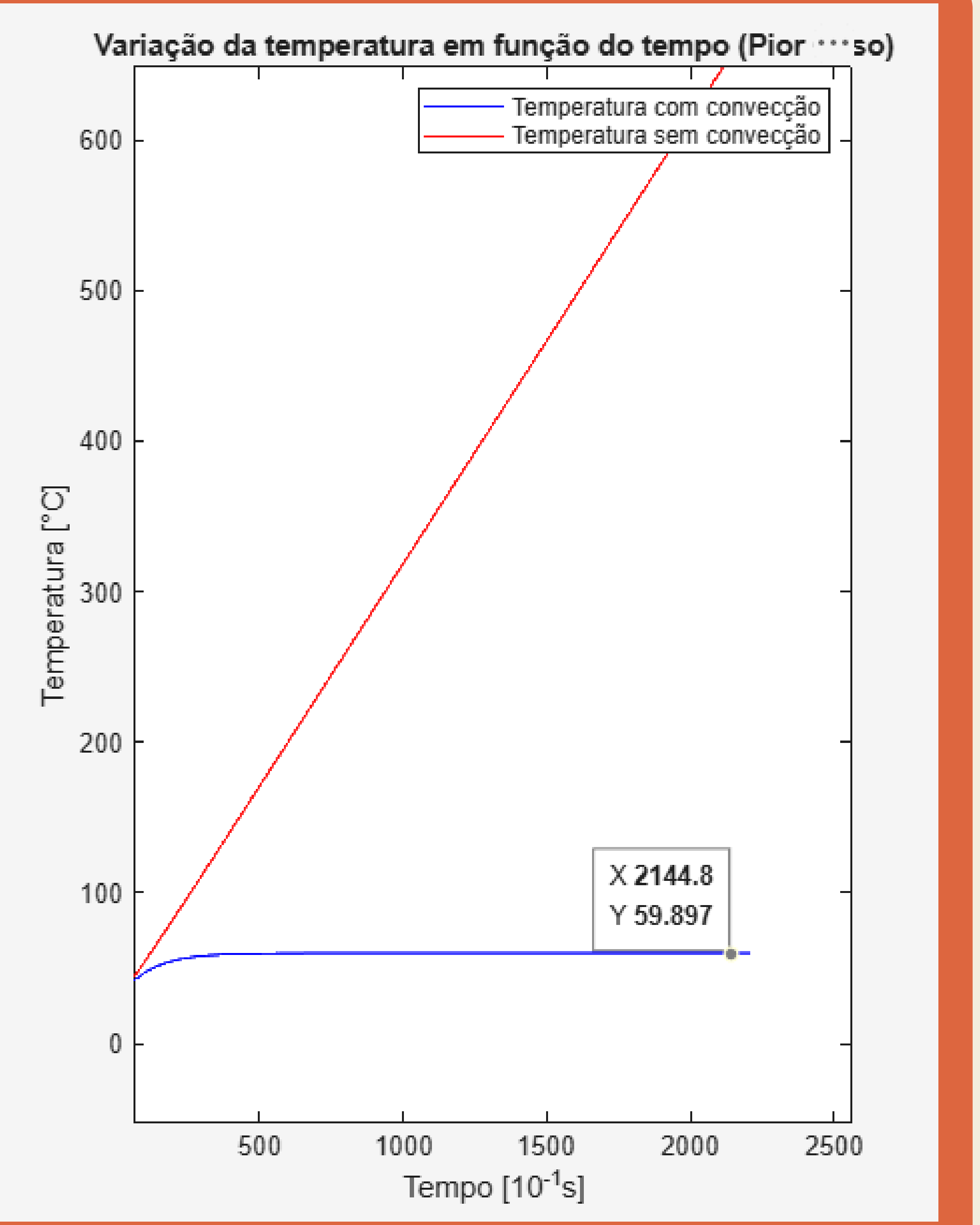


Vehicle Component Consumption Table		
Component	**Voltage (V)**	**Power (W)**
Inverter	12	21,3
VCU	12	11,3
Pump	24	133,3
GPS	12	6,6
Lidar	24	53,3
Jetson	24	106,6
Fans	24	224
Ready to Move Light	12	13,3
PCBs	12	20
Sensors	12	19,9
BMS	12	6,6
IMD	12	6,6
Brake lights	12	13,3
Access Point	24	13,3
Network switch	12	6,6



1.8. Low Voltage Battery



1.8. **Low Voltage Battery**

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Bitola AWG	Área Equivalente (mm2)	Corrente Recomendada (A)
24 AWG	0,20 mm2	Até 2 A
22 AWG	0,33 mm2	Até 3 A - 5 A
20 AWG	0,52mm2	Até 6 A - 9 A
18 AWG	0,82mm2	Até 10 A - 12 A

Grupo do Sinal	Fio	Padrão de fitas
Sinal de pré-carga	-	Azul Vermelho
CAN INS L*	Branco	Vermelho
CAN INS H*	Preto	Vermelho
Can Geral L	Branco	Azul
Can Geral H	Preto	Azul
Can Inv L	Branco	Amarelo
Can Inv H	Preto	Amarelo
Safety	-	Azul Azul
Sinais de reset	-	Amarelo Amarelo
Luzes TSAL	-	Vermelho Vermelho
Luzes falta	-	Azul Amarelo
Falta	-	Amarelo Vermelho
12V	Branco	Amarelo Amarelo Amarelo
5V	Branco	Azul Azul Azul
24V	Branco	Vermelho Vermelho Vermelho
GND	Preto	Azul Amarelo Vermelho

1.8. Low Voltage Battery